

RISK-04: REGULATORY AND LEGAL RISK DEEP DIVE

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I. EXECUTIVE SUMMARY

Regulatory risk — the risk that certification delays, adverse regulatory decisions, or changes in applicable regulations impair CSM’s ability to commercialize products — is among the most significant risk categories facing the company. This document provides deep-dive analysis of seven regulatory risks, including probability, impact, mitigation, and contingency planning.

The central regulatory reality: every product CSM sells will be installed on or connected to critical infrastructure. Every such product will require certification, approval, or acceptance by at least one regulatory body. None of these regulatory bodies currently have a specific framework for dielectric grid protection materials. CSM must both navigate existing frameworks and help build new ones. This is slow, expensive, and essential.

II. REGULATORY RISK DEEP DIVES

Risk 1: Building Code Approval Timeline Uncertainty (ICC-ES)

The Risk: ICC-ES Evaluation of CSM’s novel materials (geopolymer concrete, CMC transformer housing, MXene composite structural panels) may take 18-36 months rather than the 12-18 months projected, particularly if new Acceptance Criteria must be developed. Without an ICC-ES Evaluation Report, building officials in most jurisdictions will not approve the use of these materials in structures subject to building codes.

Probability: 4 (Likely — novel materials almost always take longer than projected) **Impact:** 4 (Major — delays revenue from any product installed in or on structures)

Detailed Mitigation: - Early engagement strategy: ICC-ES staff consultation began in Q2 2026 (pre-Series A), identifying which of CSM’s products fit existing Acceptance Criteria and which require new AC development. - Products with existing AC pathways (GridFabric panels under AC308, FRP composites): 10-16 months projected. - Products requiring new AC (geopolymer concrete for dielectric foundations): 18-36 months. This is the critical path item for building code approval. - CSM will pursue “Innovation Path” evaluation where existing AC is not available. This allows evaluation based on engineering analysis and testing without a published AC, but requires more extensive documentation and may face more scrutiny from building officials after the ESR is issued.

Testing Pre-Positioning: - Structural testing (compression, flexure, freeze-thaw, bond strength) conducted at ISO 17025-accredited laboratories before ICC-ES submission. Having completed test reports ready at submission reduces evaluation timeline by 2-4 months. - Seismic testing (shake-table per AC308) conducted in parallel with ICC-ES evaluation, not sequentially.

Contingency Options: 1. State-by-State Approval: If ICC-ES timeline extends beyond 24 months, CSM can pursue approval from individual state building code authorities under “alternate materials and methods” provisions (IBC Section 104.11). This is administratively burdensome (50 states) but faster for individual states. 2. Utility-Owned Installations: Substations owned by utilities are often exempt from local building codes (they are regulated by NERC/FERC, not local building departments). Focus initial installations on utility-owned sites where ICC-ES approval is not required. 3. Research Report Alternative: Some jurisdictions accept evaluation reports from other approved agencies (IAPMO, NSF, NTA) as alternatives to ICC-ES. CSM can pursue parallel evaluation from multiple agencies.

Risk 2: Insurance Regulatory Variation by State

The Risk: Insurance is regulated at the state level in the United States. Each state insurance commissioner must approve premium reduction frameworks. The variation in approval timelines, requirements, and willingness across

50 states creates a fragmented market adoption landscape.

Probability: 3 (Possible — insurance regulatory variation is well-documented but CSM’s approach of working through large multi-state carriers mitigates significantly) **Impact:** 3 (Moderate — slows national rollout but does not block it)

Detailed Mitigation: - NAIC Model Framework: CSM will develop a model premium reduction framework designed for NAIC (National Association of Insurance Commissioners) endorsement. NAIC model acts do not bind states but are adopted by a majority within 2-4 years. - Large Carrier Leverage: Major carriers (State Farm, Allstate, Liberty Mutual, Travelers, Chubb) operate in all 50 states and have regulatory affairs teams that manage state-by-state filing. These carriers can file the framework in all states simultaneously, achieving national coverage within 12-18 months of framework finalization. - State Sequencing: Prioritize states with (a) large commercial property insurance markets (NY, CA, FL, TX, IL), (b) favorable regulatory environments (DE, VT — captive insurance domiciles; carrier-friendly regulatory philosophy), and (c) high geomagnetic risk exposure (MN, WI, MI, NY, New England states).

Contingency Options: 1. Surplus Lines: For commercial properties, surplus lines insurers (non-admitted carriers) have greater pricing flexibility and are not subject to state rate approval. CSM can work with surplus lines carriers for faster market access. 2. Captive Insurance: Large commercial property owners with captives can adopt the premium reduction framework without state approval. Target Fortune 500 companies with captive insurers. 3. Reinsurance-Led: Reinsurers are typically not subject to state rate regulation (they are regulated in their domicile jurisdiction, often Bermuda, Switzerland, or London). Reinsurance treaty terms can require or incentivize GIC protection without state-by-state approval.

Risk 3: Export Control Restrictions on Advanced Materials

The Risk: MXene materials and EMP protection technology may be classified under ITAR or restrictive EAR categories, limiting international sales and requiring export licenses. In the worst case, technology with military applications could be subject to ITAR, which imposes a presumption of denial for exports to most countries.

Probability: 3 (Possible — the regulatory classification is genuinely uncertain, and the outcome depends on how DDTC and BIS interpret the dual-use nature of MXene composite materials) **Impact:** 4 (Major — would severely restrict international market access, which represents \$22.4B of the \$40-80B TAM)

Detailed Mitigation: - Proactive CJ/ECCN Requests: Submit Commodity Jurisdiction request to DDTC and Export Control Classification Number request to BIS in Q3 2027, seeking formal government determination rather than self-classification. This: (a) provides legal certainty, (b) creates a record of proactive compliance, and (c) allows CSM to structure products and international strategy around the official classification. - Product Architecture for Exportability: Design products so that the EMP-protection materials are separable from the end-use product. If the material itself is controlled but the product is not, international customers can purchase the product with CSM materials pre-installed, or CSM can license the material technology to in-country manufacturers. - Five Eyes Strategy: Even under the most restrictive classification, exports to Five Eyes allies (UK, Canada, Australia, New Zealand) are subject to license exceptions or streamlined review. Focus initial international expansion on Five Eyes markets, which collectively represent approximately 40% of the international TAM.

Contingency Options: 1. Foreign Manufacturing: Establish manufacturing facilities in allied countries (UK, Canada) to produce MXene composites locally under host-country export control regimes, which may be less restrictive than U.S. controls for non-U.S.-origin products (subject to ITAR “see-through” rule). 2. Licensing to Foreign Partners: License CSM technology to foreign manufacturers in target markets. Technology transfer may be subject to export control approval but a one-time license for technology transfer may be easier than ongoing product export licenses. 3. Accept Domestic Focus: If export controls severely restrict international sales, focus entirely on the U.S. market (\$18.2B utility TAM + \$6.2B defense TAM = \$24.4B). This is a large enough market to build a substantial company, though it reduces the addressable TAM and the valuation multiple.

Risk 4: Standards Committee Politics

The Risk: International and national standards committees (IEC, IEEE, NERC, ICC) are political environments where incumbents with existing technology and relationships can influence outcomes in ways that disadvantage new entrants. CSM's novel approach could be excluded from standards — or standards could be written to favor traditional approaches (Faraday cages, surge arresters) over CSM's materials-science-based approach.

Probability: 3 (Possible — standards committee dynamics are inherently political, but CSM's approach of active participation from Year 1 mitigates significantly) **Impact:** 4 (Major — if standards are written to exclude CSM's technology, the entire regulatory pathway is compromised)

Detailed Mitigation: - Active Committee Participation from Year 1: CSM budget includes \$85,000/year for standards committee participation. Viktor Kael is already establishing contact with relevant committees. The goal is not to dominate committee discussions — that would be counterproductive for a small, new entrant — but to be present, credible, and constructive. Committee members who have read CSM's technical submissions and seen the data are harder to persuade to vote against CSM's interests. - Relationship Building with Committee Leaders: Committee chairs and secretaries have significant influence over agendas, working group formation, and the pace of standards development. Building relationships with these individuals — through one-on-one meetings, collaborative research, and co-authorship of technical papers — creates channels of influence beyond formal committee voting. - Data as Persuasion: Committee decisions are made on technical grounds (ostensibly, if not always actually). CSM's strongest tool is data. When a committee is considering whether to include MXene-based protection in a standard, a NIST-verified test report showing 72 dB SE at 0.5mm thickness is more persuasive than any amount of lobbying. - U.S. National Committee Leverage: For IEC and ISO standards, CSM works through the U.S. National Committee (administered by ANSI). The USNC/IEC has significant influence on IEC standards, and CSM can influence the U.S. position through participation in USNC Technical Advisory Groups (TAGs).

Contingency Options: 1. Alternative Standards Pathway: If one standards body (e.g., IEC) adopts a standard unfavorable to CSM, pursue standards through a different body (e.g., IEEE) and advocate for recognition of that alternative standard by regulators. Multiple standards pathways often exist. 2. Proprietary Specification: If standards committees are persistently hostile, CSM can abandon the standards harmonization strategy and position products based on proprietary specifications backed by NIST and UL testing data. This is a less favorable outcome (proprietary specifications are harder to sell than consensus standards) but better than exclusion. 3. Accept and Adapt: If standards mandate traditional approaches (e.g., Faraday cage enclosures), CSM can position MXene composite as a superior way to achieve the mandated outcome (lighter, cheaper, easier to install than traditional enclosures) rather than as a replacement for the standard. Work within the standard rather than against it.

Risk 5: Product Liability Risk

The Risk: If a CSM product is installed on a transformer that subsequently fails during a geomagnetic storm — whether due to a product defect, installation error, or a storm exceeding the product's design parameters — CSM faces product liability exposure. The exposure is particularly acute because: (a) our products are sold as protection against catastrophic events, and (b) the consequences of protection failure could be severe.

Probability: 2 (Unlikely — conservative design margins, extensive testing, and QC protocols reduce the probability of defect-caused failure) **Impact:** 5 (Critical — a single high-profile failure could destroy the company's reputation and trigger costly litigation)

Detailed Mitigation: - Conservative Design Margins: All products designed with minimum 2x safety factor on critical specifications (3x for CMC transformer housing). Products are tested to 150% of rated GIC blocking capability before shipping. - Clear Specification Limits: Every product specification includes clear statements of what the product does and does not protect against. "This product protects against geomagnetically induced currents up to 100 A/phase for a duration of up to 600 seconds. It does not protect against direct lightning strikes, physical damage, or GIC exceeding 100 A/phase." This manages customer expectations and creates a clear performance standard against which liability is measured. - Installation Certification: CSM-certified installers only. Installation procedures documented and verified. QC inspection of every installation before commissioning. - Warning and Disclaimer Labels: Products carry labels clearly identifying: (a) the protection provided, (b) the protection not provided, (c) the need for regular inspection, and (d) the limitation of liability. - Product Liability Insurance: CSM will maintain product liability insurance with coverage limits appropriate to the exposure. Initial coverage: \$10M

per occurrence / \$20M aggregate. Premium estimated at \$150-250K/year (dependent on underwriting assessment of CSM's risk profile — complicated by the novelty of the product category). - Contractual Limitations: Sales contracts include limitations of liability that exclude consequential damages (lost revenue, business interruption) and cap damages at the purchase price or a multiple thereof. Enforceability varies by jurisdiction.

Contingency Options: 1. Incident Response Protocol: Pre-established incident response team and process for any product failure. Includes: immediate customer notification, on-site investigation within 48 hours, root cause analysis, corrective action, and transparent communication with affected parties. 2. Insurance-Backed Warranty: CSM may offer an extended warranty backed by a third-party insurer, transferring warranty risk from CSM's balance sheet to an insurance carrier. This is standard in the construction and industrial equipment industries.

Risk 6: Environmental Regulation of Nanomaterials

The Risk: MXenes are nanoscale materials. Regulatory agencies (EPA, ECHA in EU) are increasingly scrutinizing nanomaterials for potential environmental and health impacts. New regulations could restrict manufacturing, require expensive testing, or impose labeling or disposal requirements.

Probability: 3 (Possible — the regulatory trajectory for nanomaterials is toward increased scrutiny) **Impact:** 2 (Minor — increased compliance costs rather than prohibition of use)

Detailed Mitigation: - Proactive TSCA PMN: File Pre-Manufacture Notice (or exemption) with EPA for MXene nanosheet production before pilot plant operation begins. This establishes a clear regulatory status rather than operating in a gray area. - Environmental, Health, and Safety (EHS) Program: Comprehensive worker exposure monitoring, emission controls, waste management protocols, and community engagement. Documented compliance with OSHA PELs and EPA requirements. - NanoEHS Research Participation: CSM will participate in multi-stakeholder nanoEHS research consortia (e.g., NanoSafety Consortium, OECD Working Party on Manufactured Nanomaterials). This positions CSM as a responsible actor and provides early awareness of emerging regulatory concerns.

Contingency: MXene composite processing (extrusion, injection molding) encapsulates MXene nanosheets within a polymer matrix, preventing nanoparticle release under normal use and disposal. Even if nanomaterial regulations tighten, the encapsulated form may be exempt or subject to less stringent requirements. CSM's manufacturing processes minimize nanoparticle release through engineering controls (closed-system synthesis, HEPA filtration, wet processing).

Risk 7: Liability for Failure During an Actual Carrington Event

The Risk: This is the meta-risk — the scenario in which a Carrington Event occurs, CSM-protected assets survive but unprotected assets fail, and the owners of failed assets sue CSM on the theory that CSM's protected customers received preferential treatment or that CSM had a duty to protect all assets, not just those of paying customers.

Probability: 1 (Remote — the legal theory is novel and faces substantial defenses) **Impact:** 5 (Critical — even if CSM ultimately prevails, the litigation cost and reputational impact of being sued in the aftermath of a civilizational catastrophe could be severe)

Detailed Mitigation: - Clear Public Messaging: CSM has never claimed to protect everyone. Every public communication, from the SiblingFrequency show to investor materials, states clearly: "CSM protects paying customers who have installed our products. We cannot protect everyone. We advocate for widespread adoption, but the decision to protect rests with asset owners, regulators, and insurers." - Government Relations: Pre-event engagement with FEMA, DHS, and state emergency management agencies to establish CSM's role as a voluntary protection provider, not a government-mandated monopoly or public utility. CSM is a private company selling protective equipment to willing buyers. - Good Samaritan Doctrine: In the event of a Carrington Event, CSM will make its non-trade-secret technical information freely available to support reconstruction efforts. This is documented in CSM's emergency response plan and would be cited in any litigation to demonstrate CSM's good-faith efforts to assist beyond paying customers.

Contingency: Pre-established relationship with a law firm experienced in mass-tort and novel-liability defense. Retainer agreement for emergency legal support in the event of a Carrington Event. Litigation reserve fund (from operating cash flow or insurance).

III. REGULATORY RISK INTERACTIONS

Regulatory risks do not operate independently. The most significant risk interaction is:

Building Code Approval + Product Liability: If ICC-ES evaluation is rushed to meet a commercial timeline and results in an inadequate evaluation, the probability of a product failure in the field increases, which in turn increases product liability exposure. The mitigation is simple: do not rush. The timeline pressure is commercial, not regulatory. Accepting a delayed commercial launch is preferable to rushing certification.

Export Controls + International Revenue: If MXene materials are classified under restrictive export controls, the international TAM shrinks from \$22.4B to perhaps \$8-10B (Five Eyes allies only). This affects the financial projections and the exit multiple. The mitigation is the proactive CJ/ECCN request to establish classification certainty early.

IV. REGULATORY MONITORING AND RESPONSE SYSTEM

The regulatory environment for CSM's products will evolve over time. CSM must evolve with it. The regulatory monitoring program described in REG-05 (Compliance Framework) is the operational mechanism for staying ahead of regulatory changes.

Additionally, CSM will maintain a Regulatory Change Contingency Fund: \$250,000 set aside from Series A for unanticipated regulatory costs — new testing requirements, additional certification applications, legal fees for regulatory proceedings. The fund is replenished annually from operating revenue.

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